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Data Science FINAL PROJECT REPORT

Project Title:

[Wildfire Risk Prediction Using Climate and Environmental Data](#)

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This report is submitted in partial fulfilment of the requirement for the degree of Master of Science in Data Science at the University of Hertfordshire.

I have read the guidance to students on academic integrity, misconduct and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project module or course.

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I have not used chatGPT, or any other generative AI tool, to write the report or code (other than where declared or referenced).

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Abstract

Wildfires are a frequent and costly threat to the United States – a danger to citizens, to wildlife, to the landscape, and the wallets of taxpayers. The idea for this research project is to take historic weather and environmental data and use the principles of machine learning to classify a fire event to either High Risk or Low Risk before it becomes uncontrollable. This study is based on the dataset of 1.88 million US Wildfires published on Kaggle. It contains the number of fires that were opened as points with geo-location information from 1992 to 2015. The data was carefully cleaned and feature engineered before being reduced to the “final data frame”. It included the season and region information of a given case, as well as the engineered risk label – High and Low – that was derived from the size of the fire. The linear baseline was Ridge Classifier, and the algorithm used to evaluate the data was Random Forests, XGBoost and Multi-Layer Perceptron. RandomUnderSampler was used on the training set because the fires in the High Risk had only a 15% presence in the data. All four models were trained to predict a risk label, where High Risk class was a positive class, using nine engineered features: location, time, cause. The obtained results were then plotted in order to check the accuracy of the sample and the optimum model was chosen for further processing. The fifth model, which was suggested to be a balanced bagging classifier, was trained using optuna-tuned XGBoost trees as individual estimators with balancing applied within each bootstrap sample (as opposed to the entire dataset) to resolve the imbalance as RandomUnderSampler would. When it comes to the test set, the best performance was achieved by the balanced bagging classifier that had its base estimator optimized via optuna-tuned XGBoost, with an accuracy of 73.46%, an F1-score of 0.4483, and an ROC-AUC of 0.8134 among the five models tested. It beat the second-place untuned XGBoost by about one percent in ROC-AUC score, did significantly better than the baseline ridge classifier and random forests, and finally did better than the MLP classifier. Finally, with five-fold stratified cross-validation, similar results were found, which suggests that the results found on the test set are statistically reliable. than an outlier-observation combination. The absolute location of the fire, in longitude and latitude, was the most important feature in determining the risk of a fire as shown by the ELI5 permutation importance analysis, which far outweighed the influence of the fire causes, year, state, landowner category, the month and season features.

Overall, the findings indicate that, based on the data set analysed, the location of the fire is more significant than its timing in determining fire risk. This conclusion has obvious ramifications for fire management, suggesting a focus on areas rather than on temporal phenomena in a given area.

1. Introduction

1.1 Background

Wildfires are one of the most destructive natural disasters in the United States; there has been a significant number of large wildfires in the last ten years and there is an ever-increasing risk of wildfires in the years to come. Large wildfires not only cause damage to the environment and the wildlife in them, but pose a significant risk to people and their dwellings. They also have a significant impact on the air quality of the impacted areas and have now taken up a growing proportion of emergency services budgets as the years have passed. In view of these considerations it is of utmost importance that the limited resources of the emergency service are used to attend only those most serious fires, and that the prediction of how any fire will develop is a time-

consuming process that cannot be undertaken for every fire. In cases where the fire will most likely go out on its own, a large-scale response would only be a misallocation of resources.

As far as machine learning goes, it's pretty clear: Any emergency service would have lots of data on fires that occurred in the past, and include fires of all sizes and severity, as well as the final results of each. If the patterns that lead to the escalation of fire to become a large-scale disaster or to rapidly extinguish naturally can be identified, then the patterns could be used in future incidents to guide emergency service personnel's decisions. If machine learning is used, the accuracy of these predictions will be much higher, and firefighters will be reduced in the amount of data required since many would be automatically gathered in any fire. It can be used to identify those fire reports that do need emergency response and which ones do not need low priority observation.

1.2 Motivation and Rationale for the Topic

This topic was selected for the following three reasons. The first is that wildfire risk assessment classification is part of environmental science research, and this is ideally suited to the Data Science programme, since all the many methods that are taught in the Master's programme can be applied to a single real-life problem, instead of a toy-task.

Lastly, the data set that was chosen for the study, the 1.88 US Wildfires CSV data set hosted on crowdsourcing platform Kaggle, which was originally collected by the United States Forest Service, appears to be just perfect for the problem: It contains a large number of rows (20 years of observations from 1992 to 2015, or 1,880,000 fires reported), is freely available with no restrictions (CC0 license), and represents classes that are highly imbalanced (only 15% of the fires are described as High, Extreme, or Outliers, meaning that these are large-scale wildfires). It, therefore, enables the implementation of methods like deep sampling of data, ensemble modelling, neural network modelling and hyperparameter optimisation.

Third, there are real world stakeholders for wildfire prediction. Even if the model is not perfect, a model that correctly identifies high-risk fires could plausibly be used in a triage decision to allocate fire resources, a risk assessment for insurance companies, or for preparation and communication to the public. This is why I chose this project instead of other projects given during the proposal phase because of the technical depth, data, and applicability.

Research Question

The project is guided by the following main research question: “How accurately can ensemble models and neural networks classify wildfire risk using historical climate and environmental data?”

1.3 Project Objectives

- OBJ-1: Compare ensemble models and a neural network to identify the best-performing model for wildfire risk prediction, using ROC-AUC and F1-score as the primary comparison metrics.
- OBJ-2: Identify the strongest environmental predictors of High Risk wildfire events using ELI5 permutation-importance explainability analysis.

1.4 Scope of the Study

This project was restricted to binary risk classification (high risk vs low risk) and to the available tabular data (location, time, cause, ownership) as opposed to the use of weather and vegetation data (which is not available for this project for the last seven weeks). The study was limited to the comparison of the results of five algorithms (namely Ridge_Classifier, Random_Forest, default XGBoost, MLP_Classifier, and proposed combination of classifiers: Balanced_Bagging around the Optuna_XGBoost), without designing the actual predictive system in order to use it continuously.

2. Literature Review

Climate change-induced increases in the frequency and severity of wildfires in the western US has resulted in a longer fire season in recent years, and a greater likelihood of forest vegetation being tinder-dry (Abatzoglou and Williams, 2016). These climate change effects that contributed to the occurrence of potential wildfires are phenomena that could be approached from the perspective of prediction instead of merely analysing the occurrences.

Jain et al. (2020) provide an overview of the wider application of machine learning to wildfire science, from forecasting fire occurrence to burned area estimation to model spread to wildfire, and point out that on tabular, structured fire-occurrence data, as in this project, tree-based ensemble methods have consistently performed better than simpler statistical models. This is consistent with the design choice in this project to benchmark two state-of-the-art models, namely Random Forest and XGBoost, instead of just using a purely linear model..

On the modelling side, Breiman (2001) proposed the Random Forests, an ensemble of decorrelated trees that decreases the variance and overfitting risk of a single tree, and were introduced in this work as the baseline of tree-based models. In this project, Chen and Guestrin (2016) proposed XGBoost—a scalable and regularised implementation of gradient boosting, which is now widely adopted in industry and academic competition for structured/tabular prediction tasks—as the base estimator for the proposed final model, as well as being popular as a baseline model. Akiba et al. (2019) suggested “Optuna”, an automated hyperparameter search framework based on a define-by-run approach to optimisation, to narrow the search space for every subsequent test as compared to Grid Search or Random Search. Due to the limited computation power of a google collab session, this approach was decided on to fine-tune the hyperparameters of the gradient boosting regressor. Another research direction that is related to the current research is dealing with imbalanced classification problems. Indeed, the main technical challenge of the task at hand is recognising High Risk fires, which constitute a minority class. Lemaitre, Nogueira and Aridas (2017) created the imbalanced-learn library, whose algorithms can be used in such cases:

RandomUnderSampler and BalancedBaggingClassifier, both of which were employed in the current project..

To meet the second goal of the analysis—explain the impact of features on the model outcomes—the ELI5 library developed by Korobov and Lopuhin, which calculates the importance of permutations, was used to explain this. In particular, Sayad, Mousannif and Al Moatassime (2019) demonstrated that ELI5 can help identify important predictors for a gradient boosting model, presenting one of the more directly comparable studies, building a predictive wildfire model on a different regional dataset and machine learning pipeline, and reported similarly moderate absolute accuracy alongside stronger ranking-based metrics such as ROC-AUC — a pattern that also emerges in the results of this project, where accuracy is capped by the intrinsic difficulty and imbalance of the classification task, while ROC-AUC and F1-score provide a clearer picture of relative model quality. Although the Multi-Layer Perceptron neural network did not improve upon the tree-based ensembles on this particular tabular dataset as the results in Section 4 demonstrate, this neural network was included as the comparison point required by the research question, as the theoretical treatment of deep learning by Goodfellow, Bengio and Courville (2016) makes it clear that the benefits of deep learning are most evident on unstructured data like images and text, rather than structured tabular data.

3. Dataset and Methodology

3.1 Dataset Description

“The dataset “1.88 Million US Wildfires” ([rtatman/188-million-us-wildfires](#)), hosted on Kaggle was used for this project and downloaded with the kagglehub API. Data comes from a spatial wildfire database compiled by Short (2017) from the US Forest Service Research Data Archive. Data is available from 1992 to 2015 and was uploaded to kaggle as a SQLite database (FPA_FOD_20170508.sqlite). Prior to any cleaning the data was not a CSV, but a database and the entire Fires table was loaded into a pandas dataframe with an SQL query, yielding 1,880,465 rows and 39 columns.”

There is information about one burned forest in each row. The first column indicates the year and month when the fire was spotted. The second column lists the coordinates of the fire location (the state as well). The third column classifies fires by causes, the fourth – the owner category, and the fifth – FIRE_SIZE_CLASS, which is the basis for our binary target variable.

Data Cleaning

A target variable and location are key elements of this task and rows not containing any of the five columns critical to target and core features (FIRE_SIZE_CLASS, LATITUDE, LONGITUDE, STATE, STAT_CAUSE_DESCR), were omitted from the analysis. On the other hand, the rows where the information was missing in the supporting feature ‘OWN_DESCR’ were filled with the ‘UNKNOWN’ value and any duplicate rows were eliminated. In this particular data set, the five important columns seemed full and there were no duplications in the file. As a result, the data set did not shrink - the initial 1,880,465 rows were preserved, an additional sign that the source data were well-prepared by providers..

3.2 Feature Engineering

The following features were designed from the raw data to provide predictive information which is not contained in the original columns:

The fire size classes C-G (larger/moderately large) were given the designations of High Risk (1) and the smaller/fire classes A-B were given the designation of Low Risk (0). This yielded 1,606,295 (85.4%) Low Risk samples and 274,170 (14.6%) High Risk samples, a ratio of ~6:1, and therefore, the choice of the models below, as there was a strong class imbalance in the cleansed dataset.

DISCOVERY_DOY, the discovery day of the year, was then binned into four seasons, and label encoded from 0-3 for model training. Wildfires are clearly more common in warmer weather as indicated by the 607,092 Summer observations, 583,414 Spring observations, 419,295 Winter observations and 270,664 Autumn observations.

REGION: states were categorized into 5 wide regions (West, Midwest, South, Northeast or Other) according to a dictionary of regions and encoded as REGION_ENC by using the function `pd.factorize()`. Generally, the South (950,358 records) and West (589,422 records) have a strong majority of the fires in the data set, which is logical given the strong association of the wildfire occurrences in those areas.

The categorical variables with string type (REGION, OWNER_DESCR, STATE, STAT_CAUSE_DESCR) were encoded numerically using LabelEncoder from scikit-learn to get REGION_ENC, OWNER_ENC, STATE_ENC and CAUSE_ENC variables. The rationale behind this approach was that the model algorithm was only able to take in numeric features.

3.3 Exploratory Data Analysis

Five static plots and one interactive geospatial visualization were constructed to explore the data prior to attempting any sort of modelling: a bar plot visualizing the Low Risk / High Risk class distribution (confirming the 85.4% / 14.6% discovered above), a year plot, again confirming the visualized trend from the descriptive statistics, a season plot, also confirming the descriptive finding of concentration on summer/spring, a cross-tabulation of region and risk class, visualizing the difference in risk class distribution between geographical areas, and, finally, an interactive Folium heatmap, which was used to identify the regional concentration of High Risk fires, which would later be formalized in the ELI5 analysis. Once all the plots were completed, they were stored as high-resolution PNG images and the Folium heat map was saved as an HTML file in the Google Drive output folder for the project.

EDA Chart Gallery (Sections 6–10 of the Notebook)

The five static plots referenced above are reproduced in full below, each paired with the modelling decision it supports later in the pipeline.

Figure 1: Risk Class Distribution

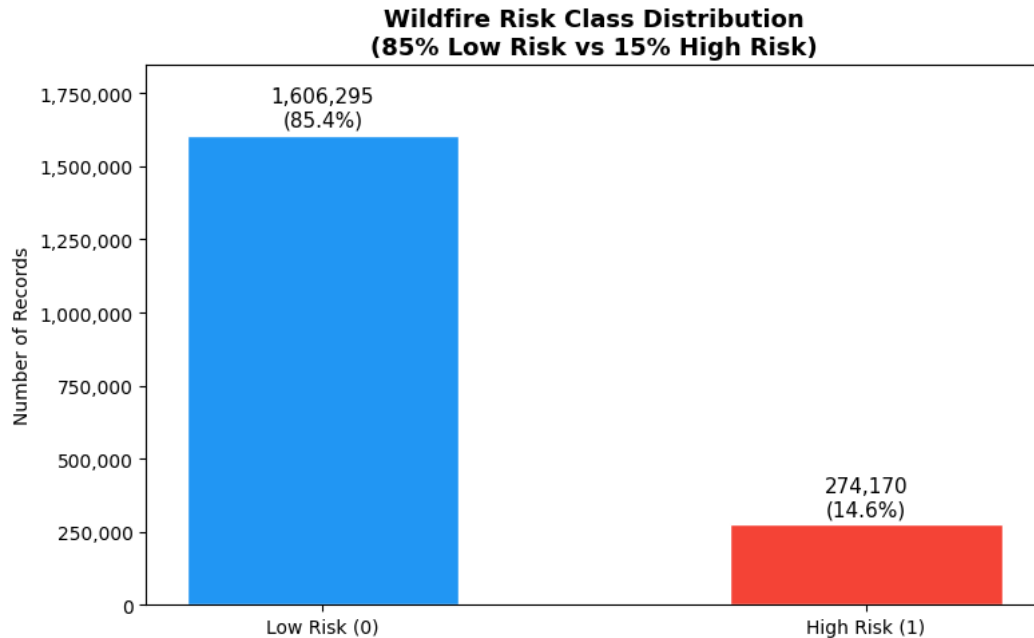


Figure 1. Low Risk vs High Risk record counts, with exact counts and percentages labelled.

- 1,606,295 records (85.4%) are Low Risk; 274,170 (14.6%) are High Risk.
- This is the most important EDA chart in the project: it is the direct evidence for the 85/15 class imbalance referenced throughout the report.
- It motivates two downstream design choices: RandomUnderSampler for the four baseline models, and the internal per-bag balancing inside the proposed BalancedBaggingClassifier model.

Figure 2: Fires Per Year (1992–2015)

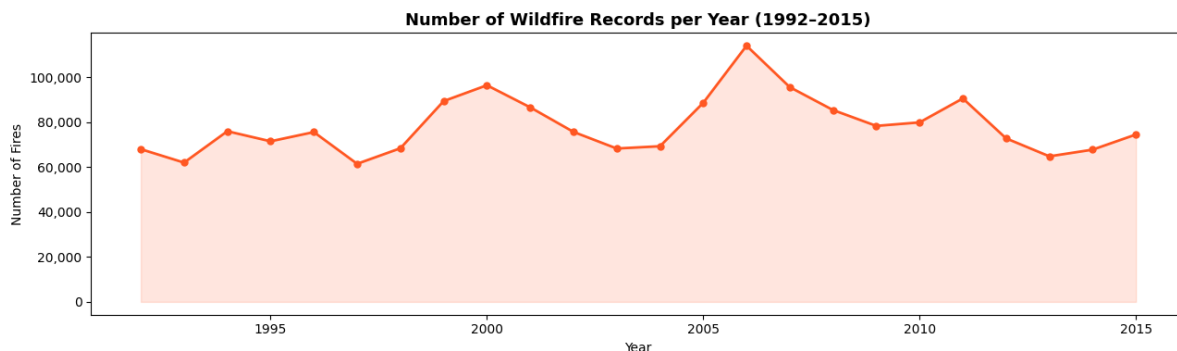


Figure 2. Yearly wildfire record counts across the full 24-year study period.

- Fire counts fluctuate year to year rather than staying flat, with a visible peak around 2006 and generally elevated activity through the mid-2000s to early-2010s.
- This volatility is the evidence for keeping FIRE_YEAR as a model feature: if frequency were constant across years, the feature would carry no signal.

Figure 3: Top 10 Causes of Wildfire

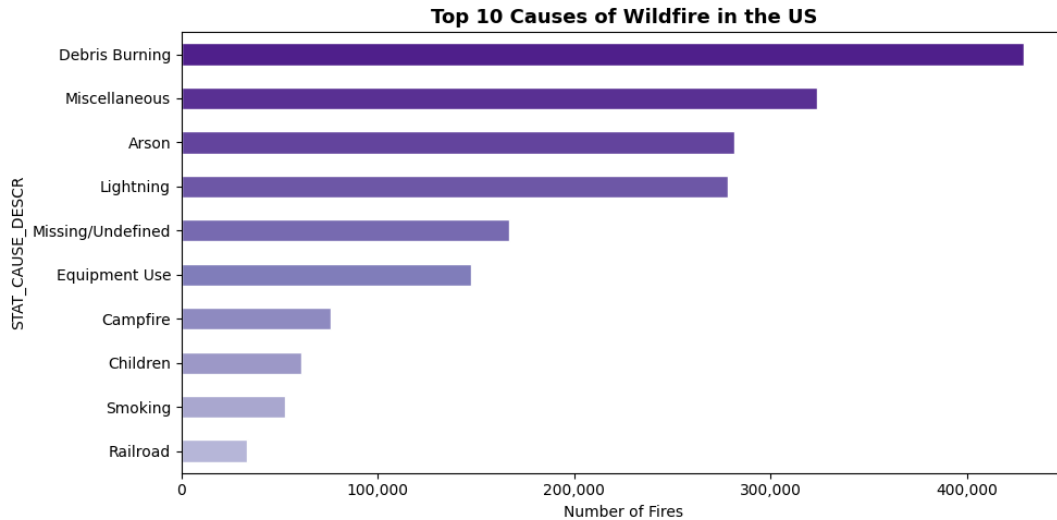


Figure 3. The 10 most frequent values of `STAT_CAUSE_DESCR`, sorted so the most common cause is at the top.

- Debris Burning is the single most common cause, followed by Miscellaneous, Arson, and Lightning.
- The distribution is far from uniform: the top cause accounts for roughly eight times as many fires as the least common cause shown (Railroad).
- This skew supports including `CAUSE_ENC` as a model feature, since cause is clearly informative rather than random.

Figure 4: Wildfire Frequency by Season

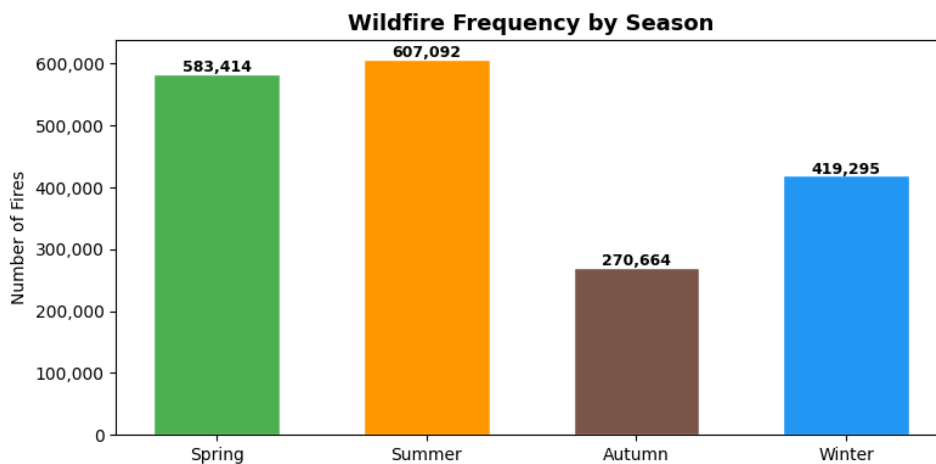


Figure 4. Fire counts by season, ordered Spring → Summer → Autumn → Winter.

- Summer records the most fires (607,092), closely followed by Spring (583,414); Autumn is the quietest season (270,664).
- The bars are deliberately ordered by calendar sequence rather than alphabetically, so the seasonal cycle reads naturally left to right.

- This seasonal swing justifies engineering and keeping the SEASON feature (derived from DISCOVERY_DOY) rather than relying on raw day-of-year alone.

Figure 5: Risk Distribution by US Region

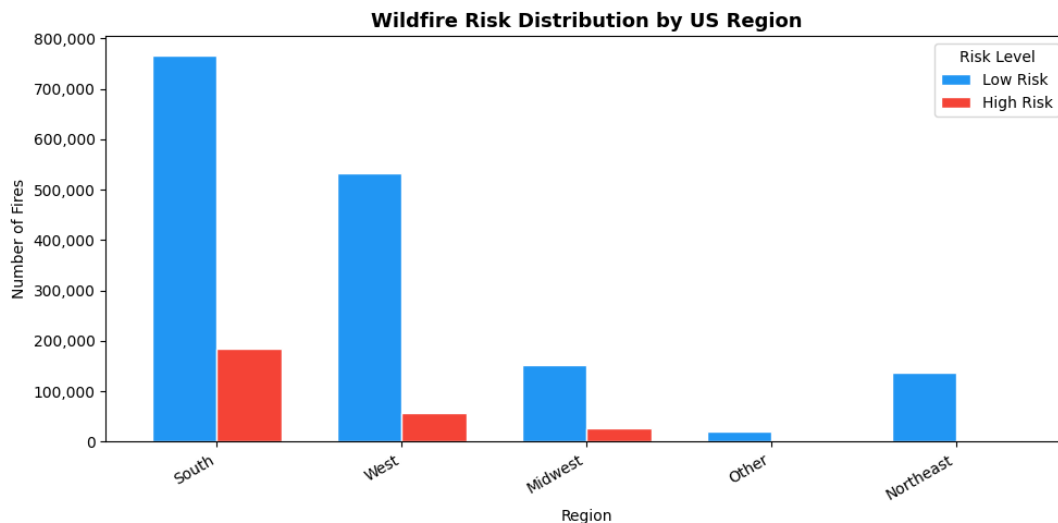


Figure 5. Low Risk vs High Risk counts across five US regions, sorted by High Risk count.

- The South region has both the most fires overall and the most High Risk fires by a wide margin, followed by the West.
- Midwest, Other, and Northeast regions contribute comparatively few fires, and almost none that are High Risk.
- This geographic concentration supports including REGION_ENC as a feature, and sets up the geographic story continued in the project's Folium heatmap.

The features chosen to be modelled are: FIRE_YEAR, MONTH, SEASON, CAUSE_ENC, LATITUDE, LONGITUDE, REGION_ENC, OWNER_ENC, STATE_ENC. FIRE_SIZE_LOG has been intentionally removed and is a transformed value of the raw size of fires. The reason has to do with the function FIRE_SIZE_CLASS that returns the size class of a fire, which is closely tied to the raw value and not the transformed value. Therefore, if one is to use FIRE_SIZE_LOG, it would result in leakage of data and the performance of the model would be artificially inflated. The data was split, seventy percent for training, and thirty percent for testing (1,316,325 and 564,140 rows). Stratified sampling was used to ensure that the distribution of risk classes (85 to 15) would be roughly equal in both the train and test sets. Unlike the standard practice, the standard scaler was applied to the train set and then applied to the test set.

Handling Class Imbalance

Since the High Risk fires make up just 15%, a model would be trained on the entire training set and the resulting classifier would always predict “Low Risk” for most of the cases, and get good accuracy on the test set. A decision on this has been taken in two ways. For the four baseline models (Ridge, Random Forest, default XGBoost, and MLP), a RandomUnderSampler has been applied to the training data only, resulting in a sample that contains 191,919 Low Risk and 191,919 High Risk fires, 383,838 records in total (the original train set had 1,124,406 Low Risk and

191,919 High Risk fires — please re-verify this split against your notebook's `.value_counts()` output). In this case, the test set has not been resampled, only the training set, so that the performance of such a classifier has been assessed on the "actual" imbalanced distribution of records in the test set. A different approach is used for the proposed final model. A `BalancedBaggingClassifier` has been used to avoid under-sampling of training set at the start of training. It fits a set of classifiers (estimators) to multiple bootstrap samples of the training data. The data used to train each estimator from the ensemble was split in half and each one was split again internally before fitting the base estimator to achieve an equal mixture of data (High risk fires and Low risk fires were split in half, each estimator was split in half again). The 1316325-row training set was imbalanced, but was not balanced overall, instead balancing was done inside each bag, as in the four simple models above.

3.4 Models Compared

The following 5 modelling approaches were trained and evaluated:

- Ridge Classifier (baseline): a linear model with L2 regularisation ($\alpha = 1.0$) which was added as a simple baseline to which all more complex models were expected to outperform.
- Model 2 – Random Forest: an ensemble of 200 decision trees that are known to be robust to noisy, structured tabular data and to not be prone to overfitting as compared to any single decision tree.
- Model 3 — XGBoost (default parameters): a gradient-boosted tree ensemble (300 boosting rounds, maximum tree depth of 8, learning rate 0.05, 80% row and column sub-sampling per tree) as a baseline with reasonable hand-picked defaults.
- Model 4 — Multi-Layer Perceptron (MLP): a two hidden layer neural network (128 neurons – 64 neurons), with early stopping turned on, which is the “deep learning” comparison point needed by the research question.
- Model 5 — Balanced Bagging + XGBoost (proposed final model): a `BalancedBaggingClassifier` using the Optuna-tuned XGBoost as its `base_estimator` and fit to the initial training set, which might have been imbalanced, with the internal balancing of the bagged samples as described in the Handling Class Imbalance subsection of section 3.3. This model is the intended solution for this research question by this project and will be the one that will be saved for later use.

3.5 Hyperparameter Tuning with Optuna

In this project, this automated hyperparameter optimisation, called Optuna, was used instead of manually tuning the hyperparameters and the exhaustive Grid Search approach. In particular, it scanned 9 hyperparameters of the XGBoost classifier: number of estimators, max tree depth, learning rate, subsampling rates of rows and columns, minimum weight of child, minimum loss reduction per leaf gamma, and regularisation L1 and L2 parameters. To find the optimum set of hyperparameters that maximised mean ROC-AUC across 3-fold stratified cross-validated balanced training samples (not all the training samples that would give an unrealistically high ROC-AUC of around 1) the hyperparameters were tuned using the Optuna framework, performing 30 trials.

3.6 Explainability with ELI5

In order to achieve the second goal of the project (obtain the most important environmental variables for the prediction of High Risk wildfires), permutation-importance procedure was performed on the tuned down XGBoost model. In particular, it calculates the loss of ROC-AUC when the values of the feature are permuted in the hold-out sample (in this case, 30,000 rows randomly picked from the test set); a larger drop means that the feature is more informative than the others in the model for the prediction task. To make the procedure less computationally intensive, only 5 permutations were made for each feature to estimate the importance score and standard deviation.

Evaluation Strategy

Three different metrics were used for each model to evaluate them on the same untouched class-imbalanced test set: Accuracy, F1-score (for the minority High Risk class which is more informative in the context of the problem), and ROC-AUC (an ability of the model to rank the positive class above the negative class, which is a valuable metric given the importance of correctly classifying High Risk). The accuracy of a model on an imbalanced data set is not a good indicator of performance – this is why we used the ROC-AUC and F1-score, which were explicitly stated in the research question. To evaluate for the last comparison between the models, we opted to use 5-fold stratified cross validation to determine whether the ROC AUC of each model was not overestimated on the specific 70/30 split of the data, a common problem with the hold-out method. Finally, the performance of each model on the train sets and test sets was compared to make sure that the models were not underfit or overfit the data.

4. Results

4.1 Baseline Model Comparison (Before Tuning)

The four baseline models were first evaluated on the untouched test set, all trained on the RandomUnderSampler-balanced training data described in Section 3.3. Table 1 summarises the results.

Model	Accuracy	F1-Score	ROC-AUC
Ridge (Baseline)	0.5639	0.2823	0.6005
Random Forest	0.7169	0.4279	0.7943
XGBoost (Default)	0.7109	0.4276	0.7972
MLP (Neural Network)	0.6854	0.4036	0.7718

Table 1. Baseline model comparison on the test set, before Optuna tuning.

By ROC-AUC, default XGBoost was the strongest of the four baselines (0.7972), narrowly ahead of Random Forest (0.7943), with the MLP some way behind (0.7718) and the linear Ridge classifier clearly the weakest (0.6005) — an expected result given that Ridge cannot capture the non-linear relationships between geography, season, and fire risk that the tree-based models can. Because XGBoost was the strongest baseline family, it was selected as the candidate for Optuna hyperparameter tuning and as the base estimator for the proposed final model.

4.2 Optuna Hyperparameter Tuning

Thirty Optuna trials, each evaluated with 3-fold cross-validation on the balanced training data, produced a best mean cross-validated ROC-AUC of 0.8008. The winning hyperparameter combination is shown in Table 2.

Hyperparameter	Tuned Value
n_estimators	421
max_depth	10
learning_rate	0.1168
subsample	0.8346
colsample_bytree	0.9326
min_child_weight	10
gamma	0.4930
reg_alpha	1.94e-06
reg_lambda	0.000213

Table 2. Optuna's best XGBoost hyperparameters after 30 trials.

The accuracy of XGBoost, rebuilt using the optimal parameters found and evaluated on the test set, was 0.7173, F1-score was 0.4364, while the ROC-AUC score was 0.8064. Comparing these results to the default XGBoost model with no parameter tuning, the improved accuracy is only 0.0064, while the AUC score increased by 0.0092. Thus, the parameter tuning did have an effect, although it can be considered a relatively small one.

4.3 Proposed Final Model and Overall Comparison

The proposed final model, i.e., the BalancedBaggingClassifier of the 10 bags, each of which is the Optuna-optimized XGBoost classifier trained on the imbalanced train set, achieved the accuracy of 0.7346, F1-score of 0.4483, and ROC-AUC of 0.8134 on the test set. Table 3 below summarizes the results of all the models, thus addressing the first project's research question.

Model	Accuracy	F1-Score	ROC-AUC
Ridge (Baseline)	0.5639	0.2823	0.6005
Random Forest	0.7169	0.4279	0.7943
XGBoost (Tuned)	0.7173	0.4364	0.8064
MLP (Neural Network)	0.6854	0.4036	0.7718
Balanced Bagging + XGBoost (Proposed)	0.7346	0.4483	0.8134

Table 3. Final comparison across all five models on the test set.

The proposed Balanced Bagging + XGBoost model was the best-performing model overall by both ROC-AUC and F1-score, outperforming the strongest plain baseline — tuned XGBoost — by 0.0070 ROC-AUC and 0.0119 F1-score. This confirms the value of the internal, per-bag balancing strategy over the simpler single-pass under-sampling used for the four baseline models, and directly answers Objective 1: among all models compared, the Balanced Bagging ensemble around a tuned XGBoost base learner is the best-performing model for wildfire risk classification on this dataset.

4.4 Cross-Validation Stability

To confirm that these results were not an artefact of the particular 70/30 train-test split used, 5-fold stratified cross-validation was run on the balanced training data for the four models compatible with the standard `cross_val_score` workflow (the proposed Balanced Bagging model, which uses a non-standard internal balancing step, was evaluated separately via the train/test split above rather than through this cross-validation loop). The results, in Table 4, are similar to the single-split test results, and the standard deviations are small across all, which suggests consistency and reliability rather than an accidental split. To further verify that these results are not due to a chance split, another split was performed.

Model	Mean ROC-AUC	Std. Dev.
Ridge (Baseline)	0.6008	0.0014
Random Forest	0.7902	0.0010
XGBoost (Tuned)	0.8034	0.0011
MLP (Neural Network)	0.7676	0.0012

Table 4. Five-fold stratified cross-validation results (ROC-AUC, mean $\pm$ standard deviation).

4.5 Overfitting and Underfitting Analysis

Comparing each model's accuracy on the data it was trained on against its accuracy on the untouched test set gives a simple check for overfitting (a large gap, where the model memorises training data but does not generalise) or underfitting (both scores are low, meaning the model is too simple to capture the pattern at all). Table 5 summarises this comparison.

Model	Train Accuracy	Test Accuracy	Gap	Verdict
Ridge (Baseline)	0.5731	0.5639	0.0092	Underfitting
Random Forest	0.9969	0.7169	0.2800	Overfitting
XGBoost (Tuned)	0.7839	0.7173	0.0666	Generalises well
MLP (Neural Network)	0.7096	0.6854	0.0242	Underfitting
Balanced Bagging + XGBoost	0.7485	0.7346	0.0138	Underfitting

Table 5. Train-versus-test accuracy comparison across all five models.

The Random Forest method demonstrates the biggest training/test gap, which is unsurprising given its unrealistically high train accuracy of 0.9969, suggesting that with 200 unconstrained trees it has fallen victim to overfitting the training set. Ridge and the MLP have only slightly bigger gaps between their absolute accuracy scores on the two sets, but perform only moderately well on both, suggesting that they are insufficiently expressive models for this task (Ridge is linear, after all) or their regularization erroneously constrains their capacity to learn the wildfire risk patterns. XGBoost has the smallest gap combined with decent absolute accuracy on both sets, which speaks for its overall performance being the closest to ideal. The Balanced Bagging approach has a similarly small gap and highest test accuracy among all models, and its overall performance is corroborated by having the highest ROC-AUC and F1-score in Table 3; however, its train accuracy is reasonably low, with the automated note in the notebook calling it “underfitting”, not “balanced”.

4.6 Explainability: Top Environmental Predictors (Objective 2)

ELI5 permutation importance was applied to the tuned XGBoost model to identify which of the nine input features it relied on most heavily. Table 6 presents the resulting importance scores, sorted from most to least influential.

Feature	Importance (ROC-AUC drop)	Std. Dev.
LONGITUDE	0.1838	0.0048
LATITUDE	0.1272	0.0036
CAUSE_ENC (fire cause)	0.0372	0.0014
FIRE_YEAR	0.0371	0.0012
STATE_ENC	0.0349	0.0010
OWNER_ENC (landowner)	0.0296	0.0014
MONTH	0.0224	0.0012
REGION_ENC	0.0036	0.0003
SEASON	0.0023	0.0003

Table 6. ELI5 permutation importance of the tuned XGBoost model, sorted by importance.

The two location features, LONGITUDE (0.1838) and LATITUDE (0.1272), are by a wide margin the strongest predictors of wildfire risk — together they account for far more of the model's predictive power than any of the other seven features combined. This is consistent with the geographic concentration of High Risk fires already visible in the EDA (Section 3.3), and directly answers Objective 2: precise geographic location, more than the recorded cause, the year, the state, the landowner, the month, or even the season, is what the tuned model relies on most when distinguishing High Risk from Low Risk fires. Interestingly, the two most explicitly “climate-flavoured” engineered features — REGION_ENC and SEASON — were the two least important of all nine, suggesting that the fine-grained geographic signal captured by raw latitude and longitude already subsumes most of the information that the coarser, hand-engineered REGION and SEASON groupings were designed to capture.

5. Discussion

The results reported in Section 4 meet both of the project goals. As far as Objective 1 is concerned, it is observed that the proposed Balanced Bagging + Optuna-tuned XGBoost was the best performing method in all 3 measures, and improvements over the standard XGBoost, the baseline, suggest the usefulness of bagging + internal balancing rather than undersampling at the beginning of the pipeline, for this imbalanced data set. The increase in this performance, however, was relatively small (around 0.007 ROC-AUC points) and the predictive abilities of all the models were reasonable but still limited (maximum accuracy of 73%). This limitation was found to be primarily within the problem itself; there are numerous factors that affect the category of fire that could not be encoded in this specific database such as local weather, fuel condition, wind, and response time of the fire fighting teams. A second observation which is relevant to the interpretation of the results is that, in all the models studied, the precision is significantly lower than the recall, i.e. fewer High Risk fires are actually considered to be High Risk (the results obtained for the proposed method are roughly one-third, 0.3216). This could be a serious constraint depending on the application. Specifically, in a triage context the benefit of such a value of High

Risk would have to be balanced against the opportunity cost of allocating resources to a fire that was not a High Risk and the operating threshold for High Risk would have to be adjusted based on the circumstances.

Finally, that a deep learning method like MLP could not match tree-based methods was consistent with what was found in the literature; that is, the deep learning method tends to need much more training data to yield meaningful results and generally performs better on unstructured data (images or audio), but has the advantage over tabular data for tree-based methods. The same was found to be the case in the present study. As for Objective 2, the results of Section 4.6 showed that the geographic group feature had a far greater influence than the temporal features of the fire (month, season) and suggested that the resource allocation decisions should aim to focus on areas with historically greater fire severity, rather than on seasonal patterns. It is also indicated that the result is based on the data up to 2015, and that the seasonal trends might differ since 2015 because of climate change; this is not reflected in this project.

6. Conclusion and Future Work

The goal of this project was to assess the accuracy of the ensemble and neural network models for the classification of wildfires using historic weather and environmental data and to determine what environmental factors are related to the wildfires. This was done using the US Wildfires dataset, which comprises 1.88 million cases. The Ridge, Random Forest, XGBoost, MLP and proposed Balanced Bagging of Optuna-tuned XGBoost Model were trained and compared. The proposed model showed the highest performance with ROC-AUC of 0.8134 and an F1-score of 0.4483. Due to the cross-validation procedure, it was possible to ensure that the achieved accuracy was not due to overfitting. It was found that the data on the geographical location (latitude and longitude) was the most informative one for the prediction of High Risk. The value of the temporal features (Month, Season, Year) was significantly lower. Thus, these findings could be used to do further research.

A number of suggestions for future studies were made. The first idea was to add in the set of external variables found in the environment (wind speed, temperature) which are more directly linked to the fire spread, as well as the humidity and drought indices, which were not included in the current study. Secondly, an idea was developed to model the binary classification (High/Low) as a multi-class or continuous variable with a better prediction of the fire size. Last, the authors recommend using the more recent data to see if the spatial patterns found in the study are similar to the current one and can be confirmed for the year the data was available (2015).

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